

7:02 AM

for images loss function is sparse_categorical_crossentropy.

Why use sparse (for images) ~~in~~ in image
when we compare there are lots of zero and to ignore those we use sparse.

Convert to find

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \\ 5 & 6 & 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 3 & 1 \\ 2 & 4 \\ 1 & 7 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

(conv 2D (-, 1, 3, 3))

$$= 1 + 0 - 2 + 1 + 0 - 1 + 2 + 0 + 0 =$$

$$\begin{bmatrix} 0 & -7 & 1 \\ 4 & 5 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Max Pooling: will only take the max number out of the result matrix (after applying kernel).

Pooling is shrinking the the matrix and taking best value pixels.

Google Teachable machine
Get started.
Vibe coding for CNN